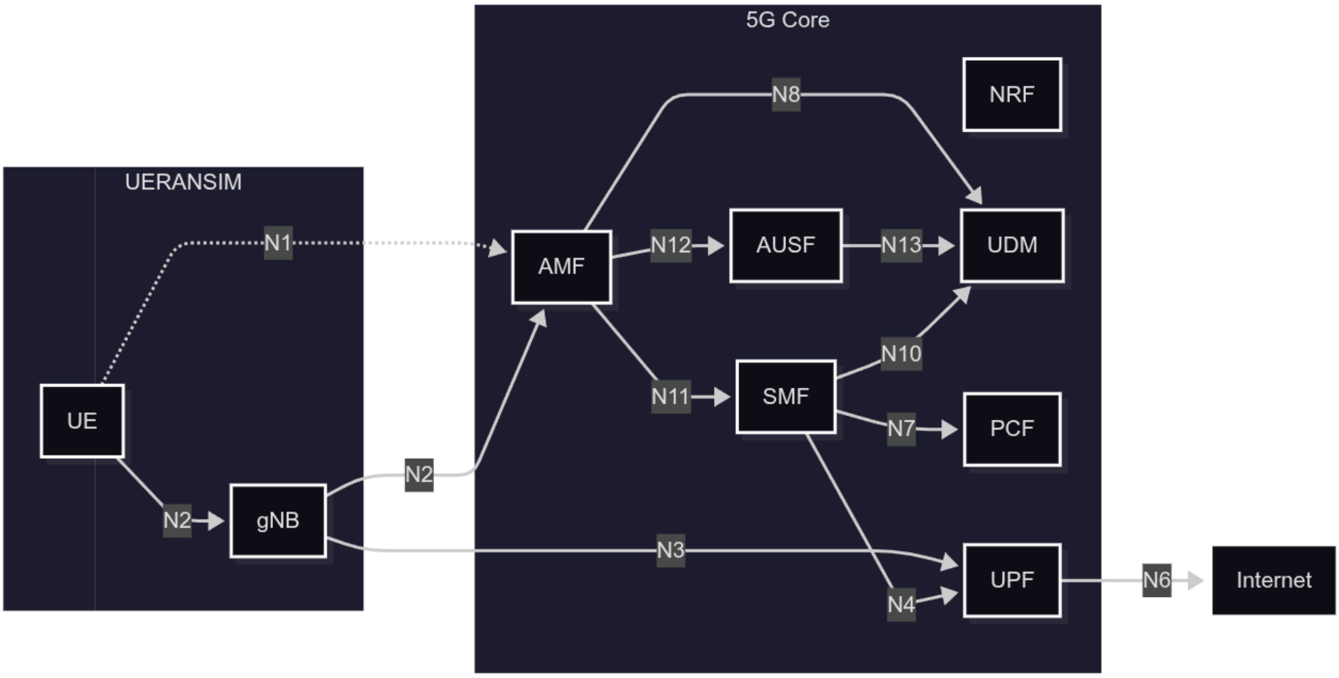
UERANSIM + Open5GS Network Performance Documentation

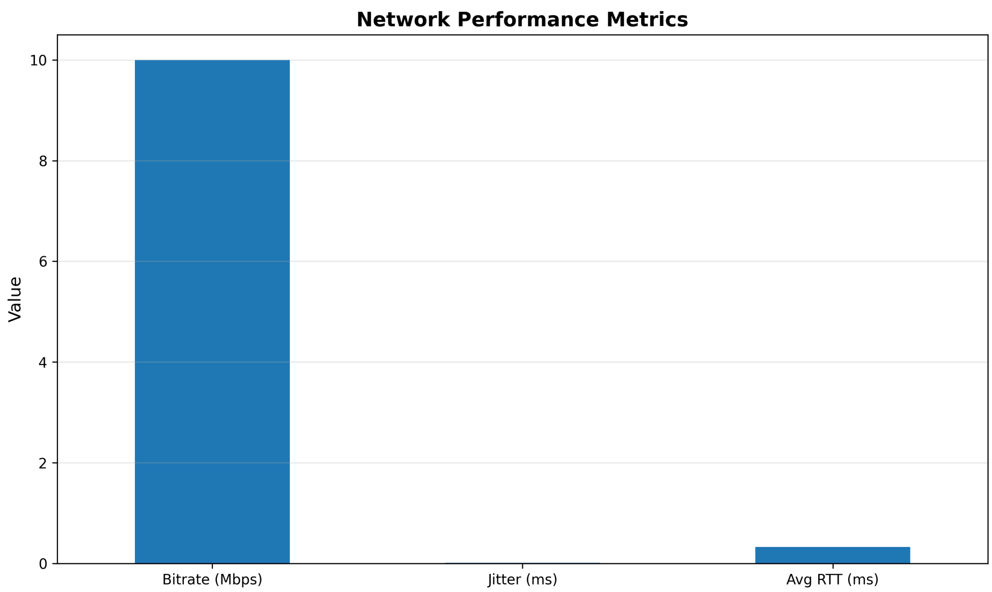
1. Working Baseline Topology + Diagram



2. Baseline Measurement Results

Performance Metrics

- **Bitrate (Mbps) - 9.99; Jitter (ms) - 0.010; Avg RTT (ms) - 0.327**



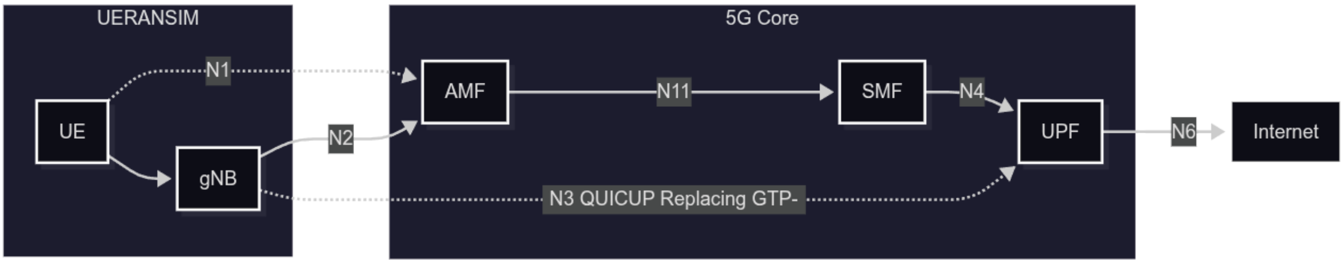
Overhead for IPv4

- GTP-U: 20(IP) + 8(UDP) + 12(GTP-U) = 40 Bytes

- GTP-U with IPSec: 20(IP)+8(UDP)+12(GTP-U)+58(IPsec) = 98 Bytes
- QUICUP: 20(IP)+8(UDP)+34(QUIC/Auth Tag)=62 Bytes

Approx. 20 bytes more for IPv6

3. QUIC-Based User-Plane Architecture



Implementation Approach:

1. **Direct Replacement**: Replace GTP-U with QUIC streams in the N3 interface
    - Simplified architecture
    - Reduced protocol overhead
    - Requires modifications to both UERANSIM and OPEN5GS
2. **Encapsulation Mode**: Encapsulate GTP-U within QUIC tunnel
    - Maintains 3GPP compatibility
    - Easier integration with existing infrastructure
    - Gradual migration path

4. QUIC Library Evaluation

Library	Language	License
quiche (Cloudflare)	Rust (C/C++ bindings)	BSD-2-Clause
msquic (Microsoft)	C	MIT
ngtcp2 (nghttp2)	C	MIT
lsquic (LiteSpeed)	C	MIT
mvfst (Meta)	C++	MIT

Recommended Implementation: **msquic**

- **Native C API**: Seamless integration with Open5GS C codebase
- **Cross-platform**: Linux compatibility for 5G deployments
- **License compatibility**: MIT license suitable for open-source projects